

Chapter 11 Checklist

- (1) Find a vector given two points.

Answer. $\vec{PQ} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$.

- (2) Determine the angle between two vectors.

Answer. $\theta = \cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} \right)$, $0 \leq \theta < \pi$.

- (3) Find the unit vector in the direction of a given vector.

Answer. Divide the magnitude, i.e., $\hat{\mathbf{u}} = \frac{\mathbf{u}}{|\mathbf{u}|}$.

- (4) Explain the relationship between the scalar component and the vector projection.

Answer. $\text{proj}_{\mathbf{v}} \mathbf{u} = (\text{scalar component}) \frac{\mathbf{v}}{|\mathbf{v}|}$, here scalar component $= |\mathbf{v}| \cos \theta$.

- (5) State the characterization of orthogonal vectors.

Answer. Two non-zero vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if $\mathbf{u} \cdot \mathbf{v} = 0$.

- (6) Find a vector that is perpendicular to two given vectors, and explain what would happen if only one vector is given.

Answer. Use the cross product. If only one vector is given, there are perpendicular vectors which form a plane perpendicular to that vector.

- (7) Calculate the area of a triangle and a parallelogram determined by vectors.

Answer. Area of parallelogram $= |\mathbf{u} \times \mathbf{v}|$; Area of triangle $= \frac{1}{2} |\mathbf{u} \times \mathbf{v}|$.

- (8) Calculate the area of a triangle given three points and a parallelogram given four points. Discuss the method on parallelogram.

Answer. For a triangle ABC , calculate $\frac{1}{2} |\vec{AB} \times \vec{AC}|$. For a parallelogram $ABCD$, calculate $|\vec{AB} \times \vec{AC}|$. It is find to calculate $|\vec{AB} \times \vec{AD}|$ as well as the result is the same.

- (9) Calculate the volume of a parallelepiped determined by vectors. Do the same if a point and its adjacent points are given.

Answer. Volume $= |\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|$. If a point A and its adjacent points B, C, D are given, form three vectors $\vec{AB}, \vec{AC}, \vec{AD}$ and use $|\vec{AB} \cdot (\vec{AC} \times \vec{AD})|$.

- (10) List the components used in describing lines and planes.

Answer. A line is described by a point and a direction vector. A plane is described by a point and a normal vector.

- (11) State the characterization of parallel lines.

Answer. Two lines are parallel if their direction vectors are scalar multiples of each other (i.e. $\mathbf{v}_1 = c\mathbf{v}_2$ for some c).

- (12) State the characterization of coplanar lines.

Answer. Given a point on each line (P_1, P_2) and their direction vectors $\mathbf{v}_1, \mathbf{v}_2$, they are coplanar if and only if $P_1 \vec{P}_2 \cdot (\mathbf{v}_1 \times \mathbf{v}_2) = 0$.

- (13) State the characterization of three coplanar vectors.

Answer. Three vectors are coplanar if and only if their scalar triple product is zero (i.e. $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = 0$).

- (14) State the characterization of parallel planes.

Answer. Two planes are parallel if their normal vectors are scalar multiples of each other (i.e. $\mathbf{n}_1 = c\mathbf{n}_2$ for some c).

(15) State the characterization of perpendicular planes.

Answer. Two planes are perpendicular if their normal vectors are orthogonal ($\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$).

(16) Find the equation of a plane containing three given points.

Answer. Form two vectors from the three points. Then find the cross product of these two vectors to get the normal vector, and use one of the points to write a plane equation.

(17) Find the equation of a plane containing two given intersecting lines.

Answer. Find the cross product of the two lines' direction vectors to get the normal vector. Use any point on the lines to write the plane equation.

(18) Find the equation of a plane containing two given parallel lines.

Answer. Choose a point P_1 on the first line and P_2 on the second line to form a vector $\vec{P_1P_2}$. Take the cross product of this vector with a line's direction vector (i.e. $\vec{P_1P_2} \times \mathbf{v}$) to get the normal vector. Use either P_1 or P_2 to write the plane equation.

(19) Find the equation of a plane containing two given skew lines.

Answer. Skew lines are neither parallel nor intersecting. It is impossible for them to lie on a same plane.

(20) Find a vector perpendicular to a given plane.

Answer. If we know the normal vector $\mathbf{n}$, we can just take $\mathbf{n}$. If the Cartesian equation $ax + by + cz = d$ is given, take $\langle a, b, c \rangle$ (which is the normal vector).

(21) Find the point of intersection of two lines.

Answer. Use different parameters (e.g., t and s) for different lines, equate x and y , solve the system for the parameters, and substitute the parameter back into line's equation.

(22) Find the point of intersection between a line and a plane.

Answer. Substitute the line's equations into the plane's equation, solve for the parameter t , and substitute t back into the line's equation.

(23) Find the line of intersection of two planes.

Answer. Find the line's direction vector using the cross product of the planes' normal vectors ($\mathbf{n}_1 \times \mathbf{n}_2$). Find a point on the line by solving the system of equations from planes.

(24) Calculate the distance between

(i) a point and a line;

Answer. $d = \frac{|\vec{PQ} \times \mathbf{v}|}{|\mathbf{v}|}$ where Q is the given point, P is a point on the line, and $\mathbf{v}$ is the direction vector.

(ii) a point and a plane;

Answer. $d = |\vec{PQ} \cdot \frac{\mathbf{n}}{|\mathbf{n}|}|$ where Q is the given point, P is a point on the plane, and $\mathbf{n}$ is the normal vector.

(iii) two parallel planes;

Answer. Choose a point on one plane and find distance between that point and the other plane, see (ii).

(iv) a line and a plane;

Answer. If they intersect, the distance is 0. If they are parallel, Choose a point on the line and find the distance between that point and the plane, see (ii).

(v) two lines.

Answer. If they are parallel, choose a point on one line and use (i). If they are skew, use $d = \frac{|\vec{P_1P_2} \cdot (\mathbf{v}_1 \times \mathbf{v}_2)|}{|\mathbf{v}_1 \times \mathbf{v}_2|}$.

(25) Calculate the angle between

(i) two lines;

Answer. Use direction vectors to find $\theta = \cos^{-1} \left(\frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{|\mathbf{v}_1||\mathbf{v}_2|} \right)$.

(ii) two planes;

Answer. Use normal vectors to find $\theta = \cos^{-1} \left(\frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{|\mathbf{n}_1||\mathbf{n}_2|} \right)$

(iii) a line and a plane.

Answer. Use direction vector and normal vector to find $\theta = \cos^{-1} \left(\frac{\mathbf{v} \cdot \mathbf{n}}{|\mathbf{v}||\mathbf{n}|} \right)$. Then add or subtract by $\pi/2$.